



Singapore's Agentic AI Governance Framework

On **22 January 2026**, Singapore's IMDA published the world's first dedicated governance framework for agentic AI. Four dimensions, one message: **as autonomy scales, so must accountability**.

What makes agents different? They plan across multiple steps, use tools, access databases, make payments, and interact with other agents. The MGF provides structured guidance for deploying these systems responsibly.

Four Levels of Human Involvement



💡 Existing governance principles still apply, but they need to be **translated into practice for agents**. Meaningful human control must be integrated into the agentic AI lifecycle.

Four Governance Dimensions

Assess & Bound the Risks Upfront

01

- Map agent action-space: tools, data, read vs. write
- Evaluate reversibility of agent actions
- Apply least-privilege access to tools & systems
- Bound autonomy with SOPs and instructions
- Establish agent identity & access controls

Make Humans Meaningfully Accountable

02

- Allocate responsibilities across agent lifecycle
- Define checkpoints for high-stakes actions
- Address automation bias with training & audits
- Clarify obligations with external vendors
- Develop adaptive governance capabilities

Implement Technical Controls & Processes

03

- Add guardrails for planning, tools & protocols
- Test task execution accuracy & policy compliance
- Whitelist MCP servers, sandbox code execution
- Gradually roll out with continuous monitoring
- Log agent behaviour and define alert thresholds

Enable End-User Responsibility

04

- Inform users of agent capabilities & data access
- Declare upfront when interacting with agents
- Train on common failure modes & oversight
- Address tradecraft erosion from agent tasks
- Provide escalation paths & human contacts

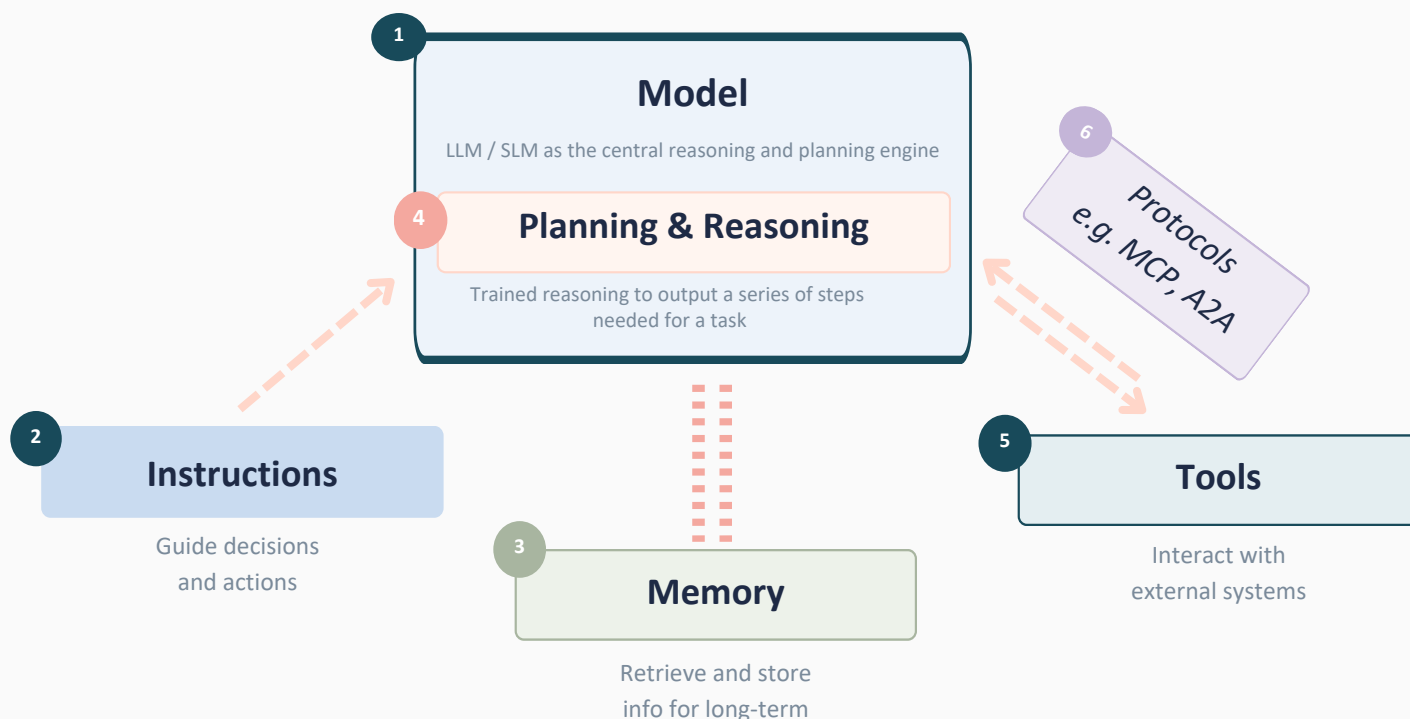
Agentic AI Risks

- ⚠ *Erroneous actions*
- ⚠ *Unauthorised actions*
- ⚠ *Data breaches*
- ⚠ *Biased decisions*
- ⚠ *System disruption*

Systemic Threats

- ◆ *Cascading failures*
- ◆ *Unpredictable outcomes*
- ◆ *Hallucinated planning*
- ◆ *Prompt injection*
- ◆ *Rogue tool use*

Core Components of an AI Agent



Arrows indicate data flow and interaction between components

Responsibility Allocation Across the Organisation

Key Decision Makers

Board, C-suite, department leaders

- Set high-level goals for agent use
- Define permitted use cases & data access limits
- Set governance approach & escalation processes

Product Teams

PMs, AI/SW Engineers, UX Designers

- Define agent design, feature controls, phased rollouts
- Implement development, testing & monitoring
- Educate users on responsible agent use

Cybersecurity Teams

CISO, Security Specialists, Pen Testers

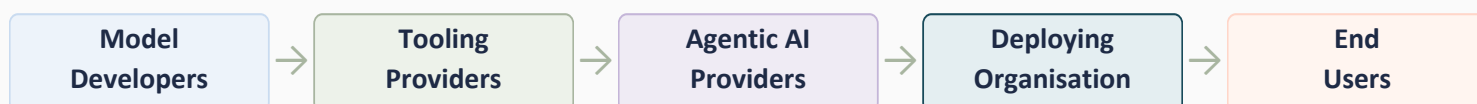
- Define baseline security guardrails & templates
- Conduct regular red teaming & threat modelling

End Users

Employees using agent outputs in workflows

- Ethical and responsible usage of agents
- Comply with policies, report bugs & issues promptly

Agentic AI Value Chain



Organisations should clarify obligations in contracts, ensure products offer security features (scoped API keys, per-agent identity tokens, tool call logging), and reassess if deployment meets risk tolerance.

This is a living document. IMDA invites feedback and case studies. Submit at: go.gov.sg/mgfagentic-feedback

Multi-agent systems require multi-layer governance.

Governance Mastery Training covers orchestration-specific frameworks and controls.

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AI Governance & Technology Policy Consulting

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